Rory Schepp

MSc Candidate in Mathematics, Dalhousie University
Email: rory@schepp.ca · Website: schepp.ca

Education

MSc, Mathematics, Dalhousie University 2025–present (expected April 2027)

- Master's thesis: Generators and relations for $O_n(\mathbb{Q})$, the group of rational orthogonal matrices. Supervisor: Dr. Neil J. Ross.

BSc (Honours), Mathematics, Dalhousie University 2021–2025

- First-class honours.
- Honours thesis: *Methods to Improve Lattice Surgery Compilation*. Supervisors: Dr. Christopher Dean and Dr. Peter Selinger.

Research interests

Group theory, including presentations of groups and matrix groups; combinatorial and computational algebra; quantum computing, including quantum circuit synthesis, compilation and fault-tolerant quantum computation; categorical and diagrammatic methods (ZX-calculus); reversible computing.

Research experience

Master's thesis research, Dalhousie University 2025–present

Supervisor: Dr. Neil J. Ross.

- Showed that $O_n(\mathbb{Q})$ is generated by 4-level matrices, i.e., matrices that act nontrivially on at most four coordinates.
- Gave a complete presentation (generators and defining relations) of $O_n(\mathbb{Q})$ in terms of n-level matrices, which act nontrivially on all n coordinates.
- Currently deriving a complete set of relations for the 4-level generators.
- Showed that $O_n(\mathbb{Q}(i))$ is generated by 2-level matrices.

Summer Research Assistant (2025), Dalhousie University Summer 2025

Supervisor: Dr. Christopher Dean.

- Developed a ZX-calculus framework for compiling quantum circuits into lattice surgery operations.
- Defined lattice surgery compatibility and showed that every Clifford+T circuit admits a compatible diagram.
- Identified ZX-calculus rewrite rules that preserve compatibility.
- Designed a procedure for embedding lattice surgery operations in three-dimensional space-time using approximate Steiner trees.

Summer Research Assistant (2024), Dalhousie University Summer 2024

Co-supervisors: Dr. Dongho Lee and Dr. Christopher Dean.

- Investigated generators and relations for reversible Boolean and ternary circuits.
- Used computational searches to explore finite presentations and identify potential relations among known generators.

Presentations

* denotes presenting author.

1. **Schepp, R.***, Dean, C. (Nov. 2025). *Methods for Improving Compilation of Quantum Circuits to Lattice Surgery.* Dalhousie Mathematics and Statistics Graduate Seminar, Halifax, NS.
2. Dean, C.*, **Schepp, R.** (Oct. 2025). *ZX-Spider Packing.* AMS Special Session on New Horizons for Quantum Error Correction: Categorical Approaches to Topological Codes, AMS Fall Eastern Virtual Sectional Meeting (online).
3. **Schepp, R.***, Dean, C. (Oct. 2025). *Methods for Improving Compilation of Quantum Circuits to Lattice Surgery.* Science Atlantic Mathematics, Statistics and Computer Science Conference, Cape Breton University, Sydney, NS.
4. **Schepp, R.*** (Oct. 2024). *Equivalence Relations on Reversible Circuits.* Science Atlantic Mathematics, Statistics and Computer Science Conference, Acadia University, Wolfville, NS.

Awards and Scholarships

2025	Science Atlantic Graduate Research Award, Third Place, Oral Presentation (Mathematics, Statistics and Computer Science Conference)
2022–2025	Dalhousie Entrance Renewable Scholarship
2021–2025	Dean's List, every term

Workshops attended

Foundational Methods in Computer Science (FMCS 2026), St. Francis Xavier University June 2026

Teaching experience

Teaching assistant and marker, Dalhousie University Sept. 2025–present

Term	Course	Role
Fall 2025, Fall 2026	MATH 1000, Calculus	Tutorial leader; problem preparation and quiz grading
Fall 2025, Fall 2026	Mathematics Learning Centre	Drop-in tutoring for first- and second-year mathematics courses
Winter 2026	MATH 1030, Linear Algebra	Tutorial leader; quiz grading and exam invigilation
Winter 2026	MATH 4116, Cryptography	Marker, weekly assignment grading
Fall 2025	MATH 3070, Number Theory	Marker, weekly assignment grading

Service

Vice-President Internal, Dalhousie Mathematics and Statistics Graduate Society Oct. 2025–Oct. 2026

Skills

Programming and tools: Python, Lean 4, LaTeX, Git/GitHub.